Set Theory Fundamentals

Lecture 1 Notes

$$\text{Probability} = \text{Set Theory} + \text{Measure}$$

1. Basic Set Operations

Union, Intersection, Complement, ...

$$A \cup B, \quad A \cap B, \quad A^c$$

2. Convergence of Real-Valued Sequences

Definition 1.1.1 — Convergence of Real-Valued Sequences

A sequence $(x_n)_{n=1}^{\infty} \subseteq \mathbb{R}$ converges to x if

$$\forall \varepsilon > 0, \ \exists N \in \mathbb{N} \text{ s.t. } n > N \implies |x_n - x| < \varepsilon.$$

We write $x_n \to x$.

You pick a threshold ε = "how close" x_n is to the target. Then for any ε picked, I can find a real value N such that for $n > N$, $|x_n - x| < \varepsilon$.

Example: the sequence $x_n = \dfrac{1}{n} \to 0$.

Definition 1.1.2 — Functional Convergence: Pointwise

A sequence of functions $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges pointwise to f if

$$\forall \varepsilon > 0, \ \forall x \in X, \ \exists N_x \in \mathbb{N} \text{ s.t. } n > N_x \implies |f_n(x) - f(x)| < \varepsilon.$$

Here $f_1, f_2, \ldots, f_n, \ldots$ are a sequence of functions, and f is the target function.

For any ε and x, we can always find N_x such that for $n > N_x$, $|f_n(x) - f(x)| < \varepsilon$ (threshold).

$$\begin{array}{lcl} f_1(1), \ f_2(1), \ \ldots, \ f_n(1), \ \ldots & \longrightarrow & f(1) \\ f_1(2), \ f_2(2), \ \ldots, \ f_n(2), \ \ldots & \longrightarrow & f(2) \end{array}$$

Definition 1.1.3 — Functional Convergence: Uniform

A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges **uniformly** to f if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ (common threshold)},\ \forall x \in X,\ n > N \implies |f_n(x) - f(x)| < \varepsilon.$$

For uniform convergence, you need a common *threshold N that works for all $x \in X$.*

Key Difference

- For **pointwise** convergence, you need to have an x specified first, then there is the threshold N_x corresponding to each x.
- For **uniform** convergence, you have the common threshold N that works for all $x \in X$.

Example: $f_n(x) = \dfrac{x}{n+x}, \quad x \geq 0.$

Then for any fixed value x, when $n \to \infty$,

$$\frac{x}{n+x} \to 0.$$

Let $x = 1$: $f_n(1) = \dfrac{1}{n+1} \to 0$ as $n \to \infty$.

Let $x = 2$: $f_n(2) = \dfrac{2}{n+2} \to 0$ as $n \to \infty$.

$$f_n(x) \xrightarrow{\text{pointwise}} 0.$$

However, it does *not* converge uniformly to 0, because we can't find an N that works for all x. No matter how you specify the value N, there's always an even larger value x.

$$\therefore f_n(x) \overset{\text{uniformly}}{\not\longrightarrow} 0.$$

3. Limiting Process

Definition 1.2.1 — The Supremum of a Set

Let $A \subseteq \mathbb{R}$ be a set. $u = \sup A$ if

$$\forall \varepsilon > 0,\ \exists x \in A,\ u - x < \varepsilon \qquad \text{and} \qquad \forall x \in A,\ x \leq u.$$

Definition 1.2.2 — The Infimum of a Set

Let $A \subseteq \mathbb{R}$ be a set. $v = \inf A$ if

$$\forall \varepsilon > 0,\ \exists x \in A,\ x - v < \varepsilon \qquad \text{and} \qquad \forall x \in A,\ x \geq v.$$

You can find $x \in A$ that's infinitely close to the upper bound u but less than it ("upper bound"); similarly v is the "lower bound."

Example: $A = (0, 1)$.

$$\sup A = 1, \qquad \inf A = 0.$$

Distinguish sup/inf **vs.** max/min

$\max(A)$ and $\min(A)$ must be elements of A, but $\sup(A)$ and $\inf(A)$ may not be.

- $A = (0, 1)$: $\max(A)$ and $\min(A)$ do not exist.
- $B = (0, 1]$: $\min(B)$ does not exist, $\max(B) = 1$, $\sup(B) = 1$, $\inf(B) = 0$.

Definition 1.2.3 — $\limsup$ and $\liminf$

Let $(x_n) \subseteq \mathbb{R}$ be a sequence.

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \left(\sup_{m\geq n} x_m \right) = \inf_{n\geq 0} \sup_{m\geq n} x_m$$

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \left(\inf_{m\geq n} x_m \right) = \sup_{n\geq 0} \inf_{m\geq n} x_m$$

$\limsup$*: the limit of the (running) upper bound.* $\liminf$*: the limit of the (running) lower bound.*

Example: $x_n = (-1)^n$, i.e. $-1, 1, -1, 1, \ldots$

$$\lim_{n\to\infty} x_n \text{ does not exist.}$$

$$\sup_{m\geq n} x_m = 1, \qquad \inf_{m\geq n} x_m = -1 \qquad \text{(specify an } n \text{ first)}$$

$$\Rightarrow \limsup_{n\to\infty} x_n = 1, \qquad \liminf_{n\to\infty} x_n = -1.$$

$\limsup$ and $\liminf$ are used to describe the long-run behavior of a sequence, even if the sequence does not converge.

For a sequence x_n: if

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = x,$$

then $\lim_{n\to\infty} x_n = x$ (the sequence x_n converges to x).

1.4 Function Limits

Definition 1.3 — Left and Right Limits of Functions

For any function $f : \mathbb{R} \to \mathbb{R}$, then:

1. $f(x^-) = \lim_{y\uparrow x} f(y) = L$

$$\iff \forall\varepsilon > 0,\ \exists\delta > 0 \text{ s.t. } \forall y \in (x - \delta, x),\ |f(y) - L| < \varepsilon.$$

2. $f(x^+) = \lim_{y \downarrow x} f(y) = U$

$$\iff \forall \varepsilon > 0,\ \exists \delta > 0 \text{ s.t. } \forall y \in (x, x+\delta),\ |f(y) - U| < \varepsilon.$$

3. If $f(x^-) = f(x^+) = C$, then $\lim_{y \to x} f(y) = C$.

Example:

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

$$f(0^-) = 0, \qquad f(0^+) = 1.$$

$$\lim_{x \to 0} f(x) \text{ does not exist.}$$